EXP. NO: 4

DATE: 10/2/26

VERIFICATION OF THEVENIN'S THEOREM

AIM :

TO Verify Thevenin's theorem practically and theoretically for the given DC circuit.

APPARATUS REQUIRED :

SI.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated power supply (RPS)	(0-30 V)	1
2.	Voltmeter	(0-30 V) MC	1
3.	Ammeter	(0-10 mA) MC	1
4.	Resistors	470 Ω, 560 Ω, 1kΩ	2, 1, 1
5.	Bread board	–	1
6.	Multimeter	–	1

PROCEDURE :

i) Make the connections as per the circuit diagram: 1

ii) Vary the RPS and set an input voltage of 10V

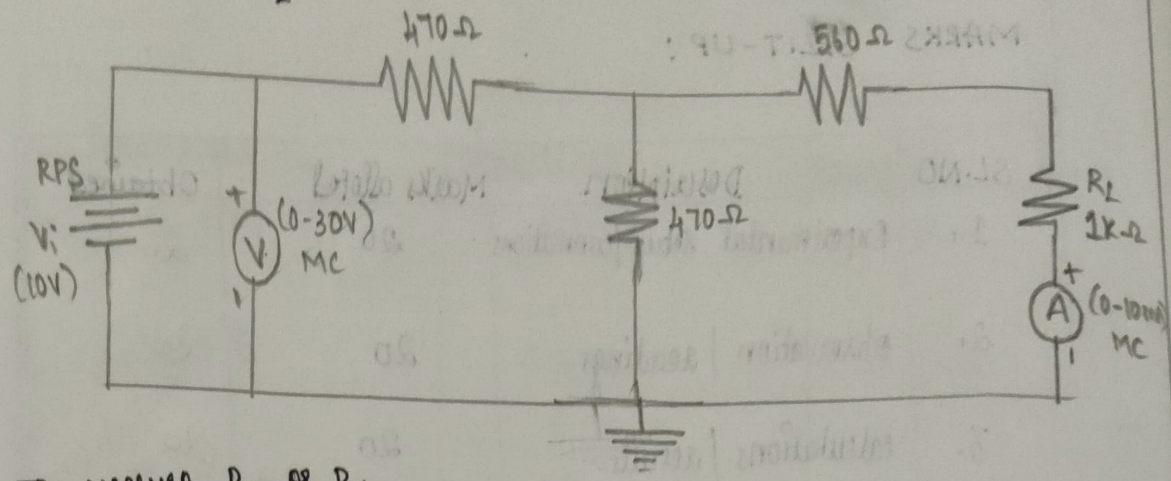
iii) Note down the voltmeter reading (V_i) and ammeter reading (I_L) in Tabular column 1.

iv) Switch off the supply and make connections for circuit diagram 2.

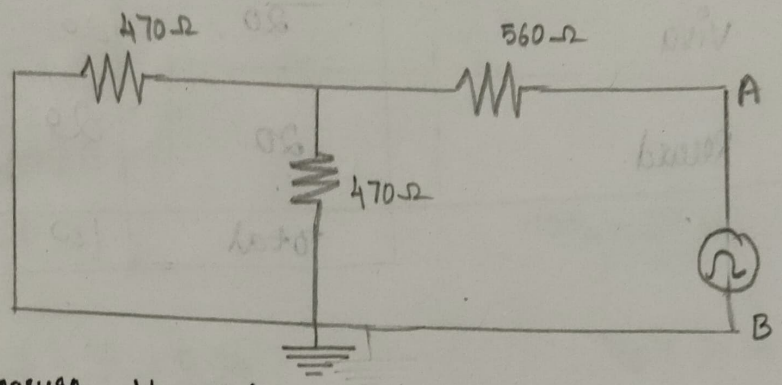
v) Measure the Thevenin's resistance R_{Th} = Norton's resistance R_N

CIRCUIT DIAGRAM :

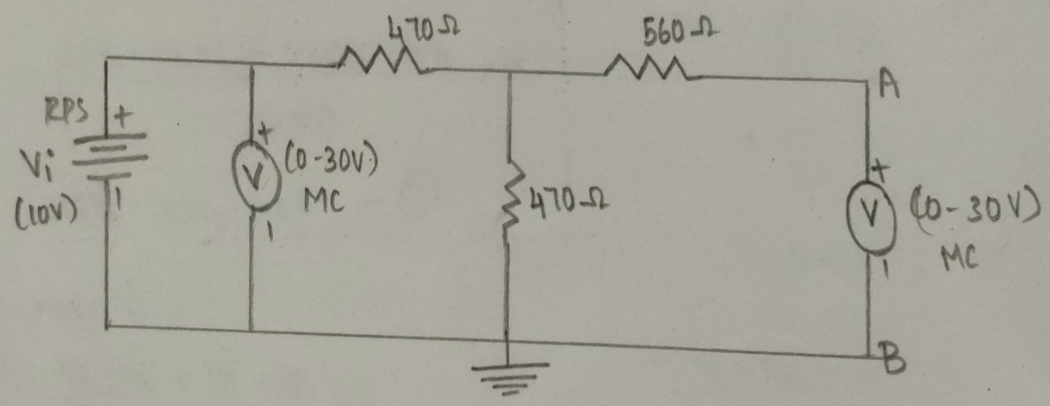
TO measure I_L .



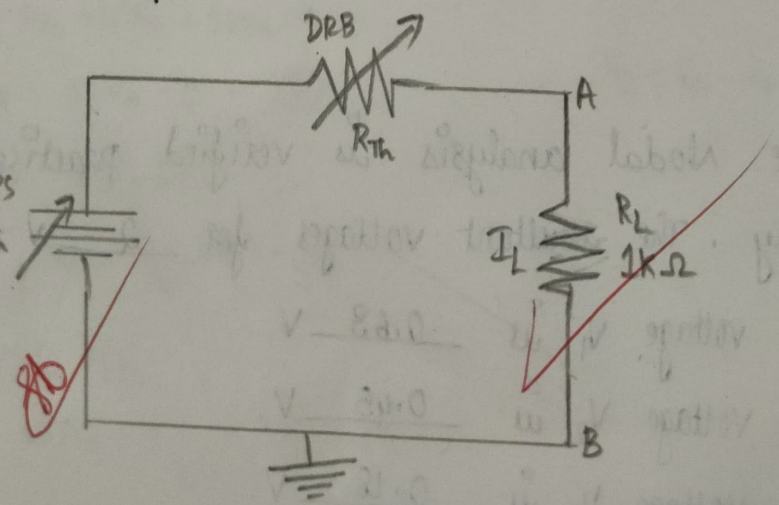
TO measure R_{Th} OR R_N



TO measure V_{Th} OR V_{OC}



Thevenin's equivalent circuit :



TABULAR COLUMN : 1

TO measure I_L

V_i (volts)	I_L (amps)
10V	2.81mA
2	0.35
4	0.71
6	1.07
8	1.43

TABULAR COLUMN : 2

TO measure R_{Th} or R_N

From circuit diagram 2, $R_{Th} = R_N = \underline{798\,\Omega}$

TABULAR COLUMN : 3

TO measure V_{Th} or V_c

V_i (volts)	V_{Th} (volts)
10V	2.8.5V

MODEL CALCULATION :

Practical value of I_L (from tabulation)

$= 2.8\,mA$

verification of Thevenin's theorem

$$I_L = \frac{V_{Th}}{(R_{Th} + R_L)}$$

$= 2.22\,mA$

vi) Switch off the supply and make connections for circuit diagram : 3.

vii) Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and V_{Th} ($= V_{oc}$) in Tabular column : 3.

viii) Switch off the supply and make connections for circuit diagram 4.

ix) Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading I_N ($= I_{sc}$) in Tabular column 4.

x) Draw the thevenin's ~~theorem~~ equivalent circuit and Norton's equivalent circuit as shown in circuit diagrams 5 & 6 respectively.

xi) Calculate the I_L value using the formula

Thevenin's Theorem

$$I_L = V_{Th} / (R_{Th} + R_L)$$

Norton's Theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

xii) Theoretically verify the Norton's theorem

THEORETICAL CALCULATION OF I_L, R_{Th} (R_N)

V_{Th} for the given circuit

By voltage division rule

$$V_{Th} = \frac{V_S \times R_S}{R_1 + R_2}$$

$$V_{Th} = \frac{10V \times 470\Omega}{(470 + 470)\Omega}$$

$\boxed{V_{Th} = 5V}$ (No current flow through R_2 when R_2 is open circuit)

R_{Th} : R_1 & R_2 is parallel when voltage source is short circuited.

$$R_{Th} = \frac{470 \times 470}{470 + 470} + 560$$

$$= 235 + 560$$

$$= 795\Omega$$

$\boxed{R_{Th} = 795\Omega}$

Find I_L :

R_{Th} in series with R_L :

$\therefore R_{eq} = R_L + R_{Th} = 1.795 K\Omega$

$$I_L = \frac{V_{Th}}{R_{eq}} = \frac{5V}{1.795 \times 10^3 \Omega} = 2.79 mA$$

$\boxed{I_L = 2.79 mA}$

MARKS SPLIT - UP :

SL.NO	Description	Marks Alloted	Marks obtained
1.	Experimental setup / connections	20	20
2.	observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total	100

100

11/2/20

RESULT :

Thus Thevenin's theorem is verified practically and theoretically.